

The Utility Problem

As with any other commodity for which it is necessary to pay, electricity is no different. In the electricity distribution business, the process of energy measurement, quantification and payment calculation is called billing. In utilities, the role of measurement is always executed by the utility meter, whose primary role is to measure and register the values which are used for energy billing.

However, due to many reasons and some specific scenarios, energy meters might not be available for the provision of sufficient data for energy billing. In these cases, utilities usually utilize alternative methods of measuring electricity for billing purposes. The most common way is the deployment of IoT grid sensors which are capable of energy measurement.

These (many times portable) IoT grid sensors offer quick and easy deployment at the desired grid location. After deployment, they substitute the role of the energy metering device, which might be missing or malfunctioning. Measured data from these IoT sensors can then be transferred to the utility billing system, or separate bills can be created using the IoT-measured data.

The Data Solutions and Methods

To allow for IoT Sensor Billing, it is necessary to have two main datasets available. First, GIS system data with consumers' locations or service point locations should be made available to accurately determine the relation of IoT sensor grid position with respect to end energy consumers. This is necessary for the reliable determination of which consumer is measured. Second, IoT-measured energy time-series data should be made available to quantify the consumed energy.

Awesense's Energy Transition Platform's data model natively supports GIS data and IoT time-series data. For users of the Awesense Platform, the IoT Sensor Billing use case can be approached in multiple ways.

The first possible way is to use Awesense's True Grid Intelligence (TGI) frontend solution. TGI's Placement page makes it possible to create a special billing placement. The TGI placement is a representation (an "envelope") for any grid measurement device, for example, an IoT device or other energy meter sensor. With the billing placement, users can populate the desired billing dates and populate required email addresses to which the IoT-measured billing data will be sent at every billing date. The IoT Sensor Billing email will contain all the necessary information for billing.

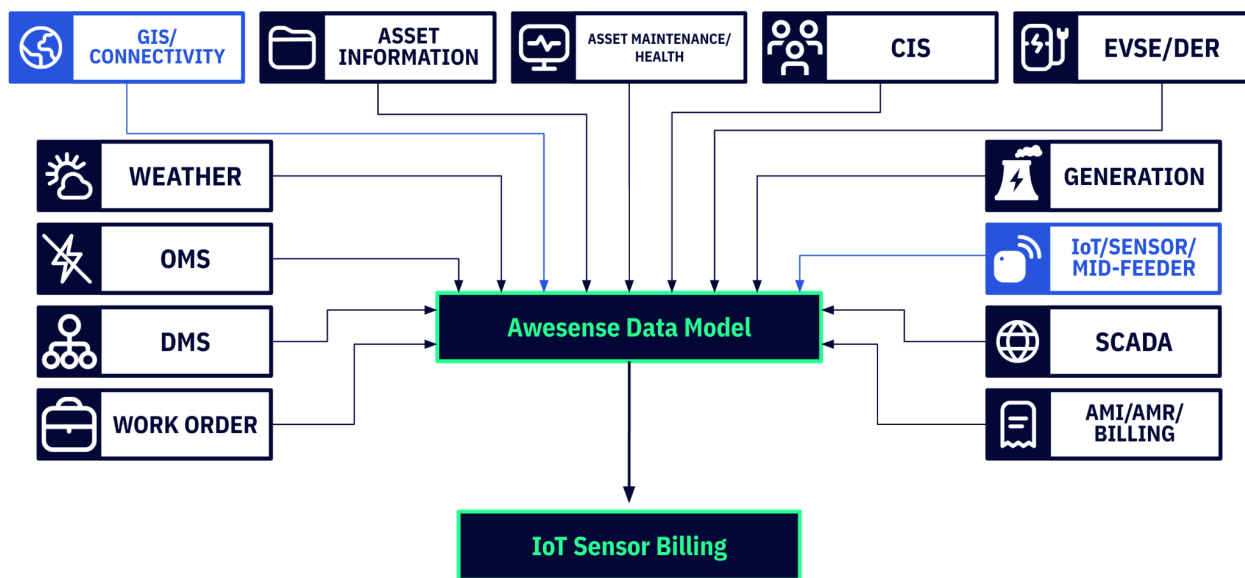
If the users would like to have more flexibility, the users can use Awesense's Energy Data Model (EDM) interfaces to programmatically interact with the data and build their own data feeds or reports to support the IoT Sensor Billing. The EDM's simplified data access methods offer this important billing-related information:

- Total energy consumption in kWh for the specified billing period;
- Peak demand (kW) for the specified billing period;
- List of measurement devices (the IoT sensors), for verification;
- Links and URLs to TGI frontend solution for further assurance;
- Many others.

With these approaches, it is ensured that utilities have a system for alternative billing using IoT Sensor with the result of no lost revenue due to missed collection of measurements.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. TGI Billing Placements - Please speak to Awesense about access to the sandbox environment.
2. A GitHub link to the Jupyter notebook can be provided upon request.